SEÁN Ó MURCHÚ

Senior Software Engineer

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Backend engineer with eight years of experience designing distributed, event-driven systems. Specialises in functional programming, observability, and developer experience.

EXPERIENCE

Senior Software Engineer

Acme Corp

Jun 2022 – Present Dublin, Ireland

Platform team lead. Owns the event-sourcing stack underpinning the wider product surface.

- Led the migration of a customer-facing monolith into a set of event-driven microservices, halving p99 latency.
- Designed and rolled out an event-sourcing platform now used by four product teams.
- Mentored three engineers from mid to senior level.

Software Engineer

Liffey Labs

Jun 2019 – Jun 2021 Remote

- Built and shipped the first version of a SaaS product from scratch alongside a two-person team.
- Introduced the CI/CD pipeline and developer-onboarding flow that scaled the engineering org from 3 to 15.

VOLUNTEER

Mentor

CoderDojo Dublin

Jun 2020 – Present

Volunteer-led coding club introducing young people to programming.

- Weekly mentoring sessions for 10–14 year-olds learning to code.
- Curate the Scala / functional-programming track.

PROJECTS

Application

Hyperion

Acme Corp

Lead Architect

Jan 2024 – Aug 2024

Distributed task scheduler in Rust

- Handled 10k tasks/s on a single node.
- Designed an idempotent retry protocol.

Rust Tokio gRPC

AREAS OF FOCUS

- Distributed systems: Kafka, CDC, idempotent consumers, event sourcing.
- Functional programming and type-driven design in Scala and Haskell.
- Developer experience: tooling, CI/CD, observability.
- Engineering culture: ADRs, RFCs, mentoring.

SKILLS

Languages – Scala Haskell Python Go Rust

Infra – Kafka AWS Terraform Docker Kubernetes

LANGUAGES

English

Irish

French

Portuguese

EDUCATION

M.Sc. in Computer Science

Tallaght Institute of Technology

2017 – 2019

Distinction

Distributed Systems Type Theory Concurrency

Computer Science

Trinity College Dublin

2014 – 2017

Crucible

Maintainer

Sep 2024 – Present

Migration tool for legacy databases

- Schema diffing across PostgreSQL and MySQL.

Library

Tinkerbell

Tiny IRC bot

Lua

Other

Quill

PUBLICATIONS

Articles

- *Event Sourcing in Practice*.
  Personal Blog · 14 Jun 2024
  A walk-through of idempotent Kafka consumers and the replay strategy that ships with them.
- *Untyped Note*.
  Jan 2024

Conference Papers

- *Idempotent Kafka Consumers at Scale*.
  EuroSys · Apr 2024

Talks

- *Designing for Failure*.
  ScalaIO · Sep 2023

Books

- *Functional Domain Modelling*.
  Self-published · 2022

INTERESTS

Music – Trad Jazz

Sport – Hurling Climbing

Travel – Japan Iceland

CERTIFICATIONS

Certified Kubernetes Administrator

Aug 2023 CNCF

Certified Kubernetes Application Developer

Jan 2024 CNCF

AWS Solutions Architect — Professional

May 2023 AWS

AWS DevOps Engineer — Professional

Sep 2024 AWS

Hashicorp Terraform Associate

Dec 2022 Hashicorp

Certified Kubernetes Security Specialist

Mar 2024 Linux Foundation

Google Cloud Professional Architect

Nov 2023 Google Cloud

AWARDS

Best Paper Award

ScalaConf

Sep 2024

For Idempotent Kafka Consumers.

Dean's List

REFERENCES

Dr Ada Lovelace

Seán is a calm, methodical engineer who left every system he touched in a better state than he found it. Recommended without reservation.

Pragmatic, curious, unfailingly kind — would hire again in a heartbeat.